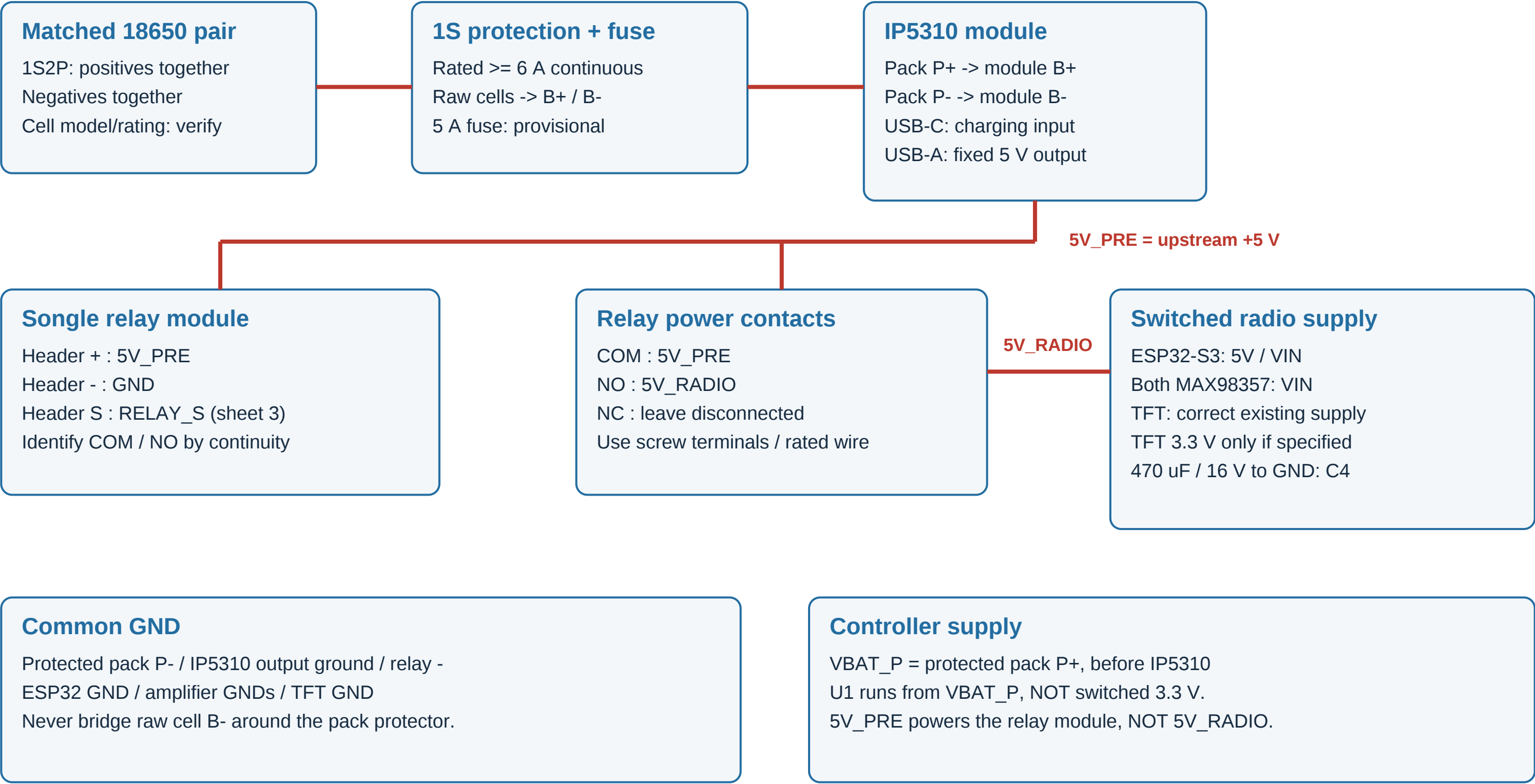


01 / Power wiring

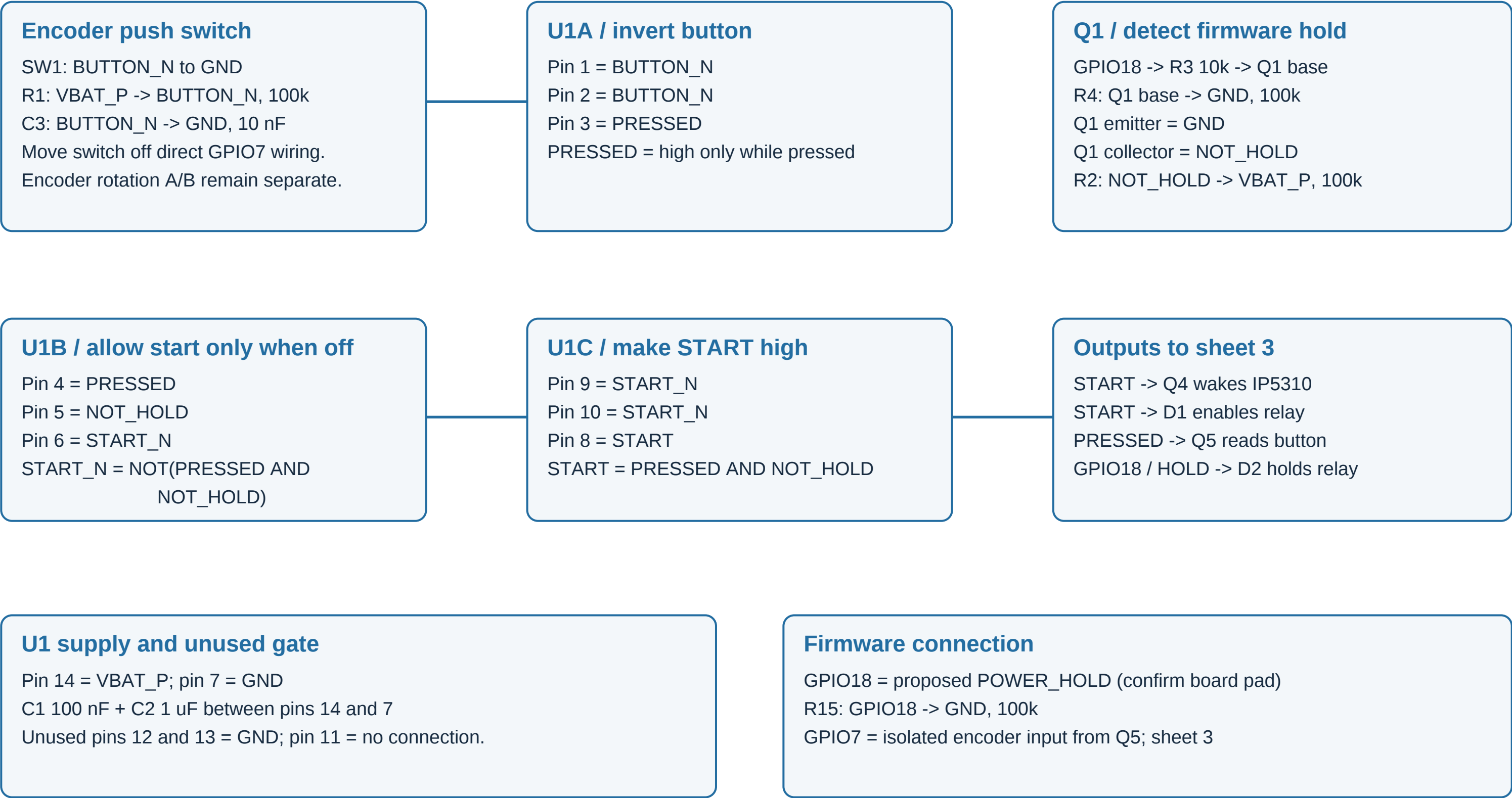
Existing relay + IP5310 charger/boost board + a small start/hold interface



No TP4056 in parallel with this charger. No programming USB cable during battery operation.
Keep speaker terminals on amplifier +/- outputs; neither speaker terminal goes to GND.

02 / Encoder and start logic

U1 = SN74HC132N, PDIP-14; named nets connect between sheets

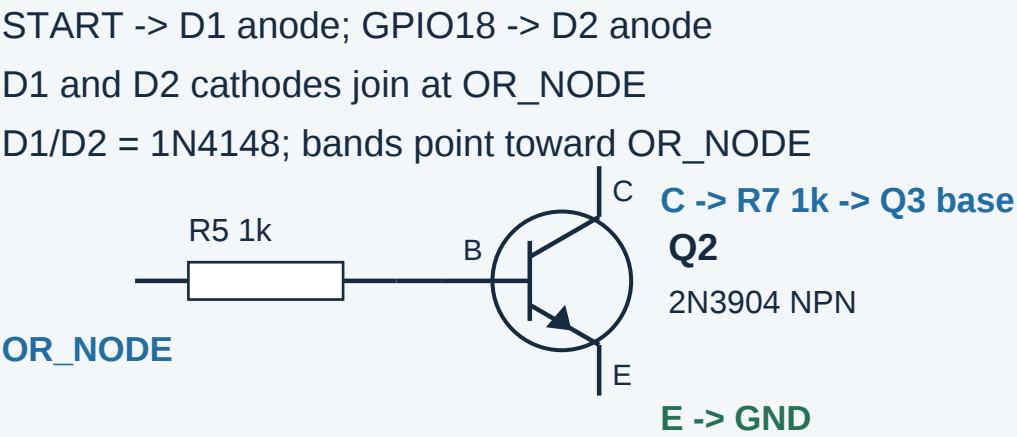


GPIO18 is unused in the inspected firmware pin map; board exposure still needs confirmation.
No battery-voltage logic signal is wired directly into an ESP32 input.

03 / Relay driver and isolated interfaces

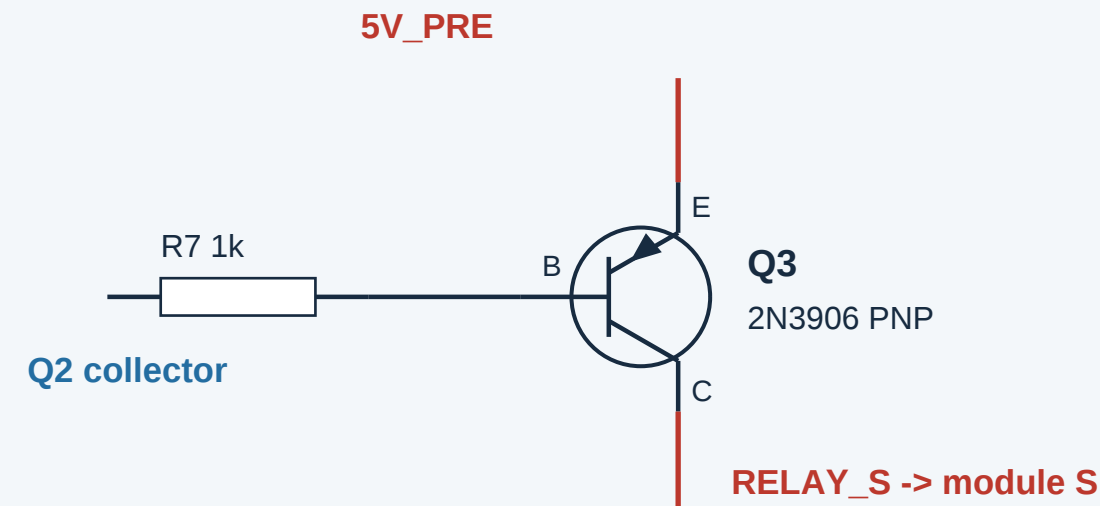
All resistors are 1/4 W. Diode band = cathode. All NPN emitters connect to GND.

Relay driver / active-HIGH module only



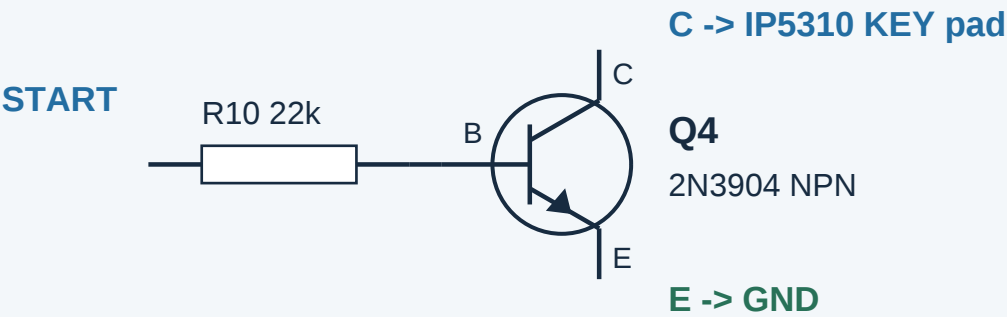
R6: Q2 base -> GND, 100k
Q3 PNP: emitter = 5V_PRE; collector = RELAY_S
R8: Q3 base -> 5V_PRE, 47k; R9: RELAY_S -> GND, 10k

Q3 / supply 5 V to relay signal



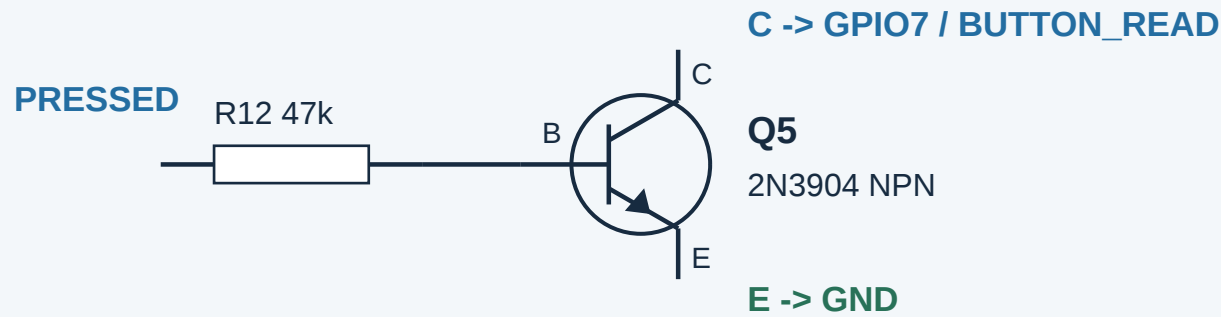
Driver intended for S input current <= 30 mA.
First verify that HIGH turns your module ON.

Q4 / wake IP5310 without GPIO backfeed



R11: Q4 base -> GND, 100k
KEY ground pad joins common GND. Keep onboard button.

Q5 / encoder input stays at 3.3 V



R13: Q5 base -> GND, 100k
R14: GPIO7 -> switched ESP32 3.3 V, 10k

Q1/Q2/Q4/Q5 = 2N3904. Q3 = 2N3906. Use E/B/C from the actual purchased transistor datasheet.
The existing relay module already contains the coil driver and flyback diode.

04 / Pin map, bill of materials and state table

The accompanying connections.csv lists both endpoints of every added component.

U1 top view / notch upward

1	BUTTON_N	14	VBAT_P
2	BUTTON_N	13	GND
3	PRESSED	12	GND
4	PRESSED	11	unused
5	NOT_HOLD	10	START_N
6	START_N	9	START_N
7	GND	8	START

Additional parts

1 x SN74HC132N + DIP-14 socket
4 x 2N3904 (Q1,Q2,Q4,Q5)
1 x 2N3906 (Q3)
2 x 1N4148 (D1,D2)
7 x 100k resistors
3 x 10k resistors
2 x 1k resistors
2 x 47k resistors
1 x 22k resistor
100 nF, 1 uF, 10 nF capacitors
470 uF / 16 V bulk capacitor
Perfboard / connectors / rated wire

Existing / pack hardware

Your Songle 5 V relay module
IP5310 USB-C + USB-A board
Two matched 18650 cells
1S pack protection >= 6 A
5 A pack fuse: provisional
5 V USB supply with load/charge margin
Do not carry radio power through a solderless breadboard.
Use bench supply before real cells.

Logic state table			
BUTTON PRESSED	HOLD	START / KEY pull-down	RELAY
No	Low / unpowered	Off	Off
Yes	Low	On	On while pressed
No	High	Off	On
Yes	High	Off	On; GPIO7 reads press

GND always means protected pack P-. VBAT_P means protected pack P+.
Confirm protection and fuse ratings against the actual cells, enclosure wiring and measured load.

05 / Commissioning and firmware handoff

Prototype acceptance steps - firmware has NOT been changed or flashed.

Before connecting the radio

- 1. Relay only: use a current-limited 5 V bench supply.
 - 2. Header + to 5 V, - to GND. S to GND: relay must OFF.
 - 3. S to +5 V: relay must ON; measure S current ≤ 30 mA.
 - 4. Identify COM/NO/NC by continuity; mark the terminals.
 - 5. IP5310: identify KEY and ground button pads by continuity.
 - 6. Confirm B- and output ground topology on the exact board.
 - 7. Verify 5 V output and 4.2 V battery configuration.
 - 8. Assemble controller; test with a dummy load first.
- If the relay is active-LOW, this drawing must be revised.

Firmware changes required before use

Early setup: GPIO18 OUTPUT LOW.
While start button is held, allow ≥ 400 ms for KEY wake.
Then set GPIO18 HIGH before releasing start button.
Consume the power-on press; do not enter AP/calibration.
Normal GPIO7 input: pressed LOW; released HIGH.
Shutdown: wait for release, save state, stop audio.
Set GPIO18 LOW as the final action. No deep-sleep handoff.
During flash/OTA writing, defer power-off.
A reset/Hi-Z HOLD drops the relay in this simple design.

User operation

OFF -> hold encoder about 1 second, until takeover.
ON -> short clicks mute; the IP5310 KEY stays inactive.
OFF request -> release encoder before power is cut.
USB connection may wake IP5310, but must not close relay.
Press held during shutdown would otherwise keep START on.
Boot recovery needs a separate intentional gesture.

Required bench evidence

Test battery-only and USB-connected operation.
Measure relay start/hold at low battery, full battery and USB.
Verify KEY low interval ≥ 300 ms on battery startup.
Verify no backfeed at GPIO7/GPIO18 while ESP32 is off.
Test fast repeated clicks, held button and aborted startup.
Measure whole-pack OFF current after boost enters standby.
Test loud playback and charge/load transfer for resets.

Sources: TI SN74HC132 datasheet; onsemi 2N3904/2N3906; Injoinic IP5310 V1.37; Songle SRD datasheet.
This is a schematic plan, not a Fritzing .fzz project, PCB layout, SPICE proof or hardware certification.
Supply/cell markings and module input behavior remain acceptance dependencies; see README.md.